(a) Conceptual Framework of the Integrative Review

1. RESEARCH QUESTIONS (SANRA GUIDELINES)

- **RQ1 :** Planetary surface magnetic vector magnitude & lander/payload superposition
- **RQ2 :** First 5 days seedling kinetics: germination capacity, emergence lag (t_{50}), biomass
- **RQ3 :** Quantum cryptochrome radical pairs & multi-stressor ROS bioenergetic nexus
- **RQ4 :** Ground simulation standards, hardware self-magnetic mapping & flight decision gates

2. MULTI-DISCIPLINARY DATA SOURCES & REPOSITORIES

- **Planetary Geophysics:** Lunar Prospector LP-MAG, Kaguya LMAG & Apollo vector models (NASA PDS / JAXA)
- **Plant Space Biology:** Peer-reviewed literature across PubMed, Web of Science, Scopus, and Europe PMC
- **Quantum Biology:** Quantum Biology Atlas (QBM-06) enzymatic & cryptochrome spin-chemical models
- **Spaceflight Multi-Omics:** NASA OSDR (OSD-782, OSD-8, OSD-27) & Astronaut Multi-Omics Consortium profiles

3. SELECTION & SCOPING CRITERIA

[INCLUSION CRITERIA]	[EXCLUSION CRITERIA]
(+) Near-null field exposure ($\|\mathbf{B}\| < 5\ \mu$T)	(-) High static fields ($\|\mathbf{B}\| > 1$ mT, e.g. MRI)
(+) 0–5 day seedling developmental stages	(-) Post-Day 14 whole-life without early data
(+) Empirical kinetics, pigment, or omics data	(-) Animal-only models without bioenergetics
(+) Calibrated planetary & orbital datasets	(-) Unshielded controls lacking ambient logs

4. MULTI-SCALE THEMATIC SYNTHESIS (5 CORE DOMAINS)

- **D1: Planetary Environment:** Ambient lunar South Pole null void (<0.82 nT) & spacecraft superposition
- **D2: Seedling Kinetics:** First 5 days: germination lag, +16 h emergence lag, -28.5% Day 5 biomass
- **D3: Quantum Biology:** Cryptochrome radical pairs, spin-chemical shifts & mitochondrial electron leak
- **D4: Multi-Stressor Nexus:** Exploration atmosphere (elevated CO2, 34% O2) & radiation overlap (AOX1D)
- **D5: Risk Retirement:** 3-chamber simulation standards, hardware power mapping & flight decision gates

(b) Evidence Identification & Selection Flow (PRISMA/SANRA)

IDENTIFY

DATABASE RECORDS
PubMed: n = 542
Web of Science: n = 418
Scopus / Europe PMC: n = 286
NASA OSDR Queries: n = 182

Total Records: n = 1,428

PUBLIC DATASETS
NASA OSDR: OSD-782, 8, 27 (n=3)
NASA PDS / SELENE Grid (n=1)
Quantum Biology Atlas QBM-06 (n=1)
Astronaut Multi-Omics (n=1)

Total Datasets: n = 6

SCREEN

RECORDS SCREENED
Duplicates Removed:
n = 512 records

Title & Abstract Screening:
n = 916 unique records

RECORDS EXCLUDED (n = 701)

- Static high-field / MRI (>1 mT): n = 314
- Animal / clinical (no overlap): n = 228
- Whole-life lacking 0–5 d: n = 159

ELIGIBLE

FULL-TEXT ASSESSED

Full-Text Articles Assessed:
n = 215 articles

Evaluated against magnetic shielding & 0–5 d phenotypes

FULL-TEXT EXCLUDED (n = 135)

- Missing magnetic field logs: n = 64
- Incomplete gas/light metrics: n = 43
- Unshielded DC confounders: n = 28

INCLUDED

SYNTHESIZED REVIEW CORPUS & COMPUTATIONAL ARTIFACTS

80 AUDITED STUDIES

Curated in Supplemental Table S1
Detailed by author, year, field (nT), model species, and biological endpoints

6 PRIMARY DATASETS

Curated in Supplemental Table S2
NASA OSDR-782, OSD-8, OSD-27; PDS/SELENE; QBM-06; Astronaut Omics

Open-Source Analytical Package: dr-richard-barker/NMF_review (GitHub & Zenodo)